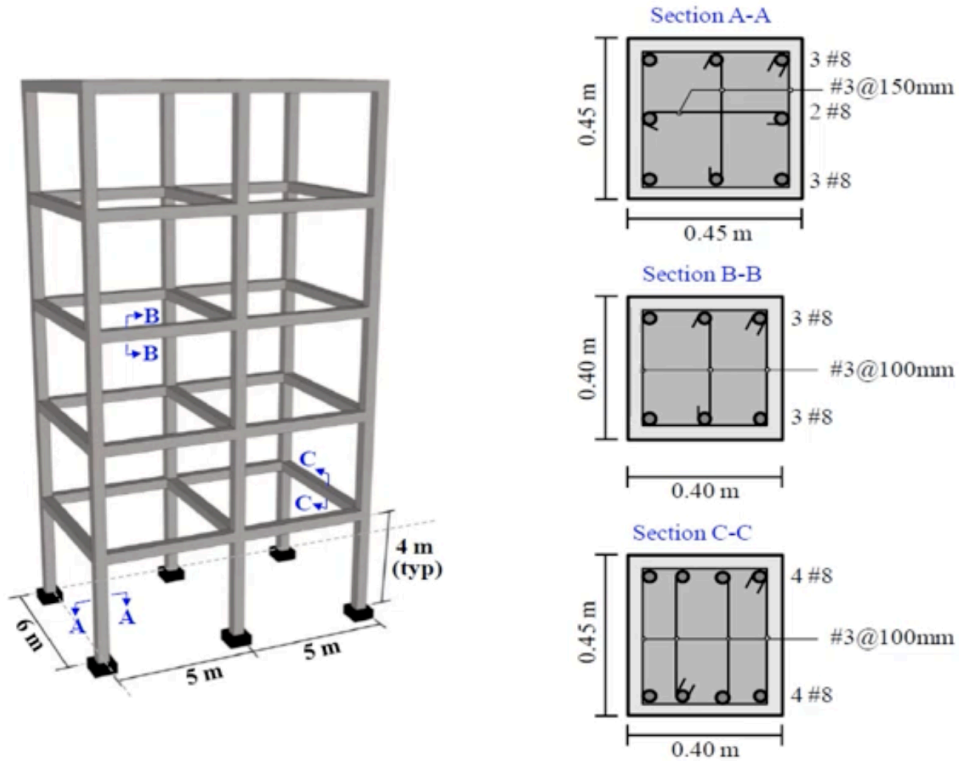


Session 1 Assignment (Full Frame Analysis)

Linear/Nonlinear Analysis of a 3D Frame

The geometrical properties of a building frame structure are shown in Figure 1.



For All Floor,
Dead Load = 6.5 kN/m^2
Live Load = 2.5 kN/m^2

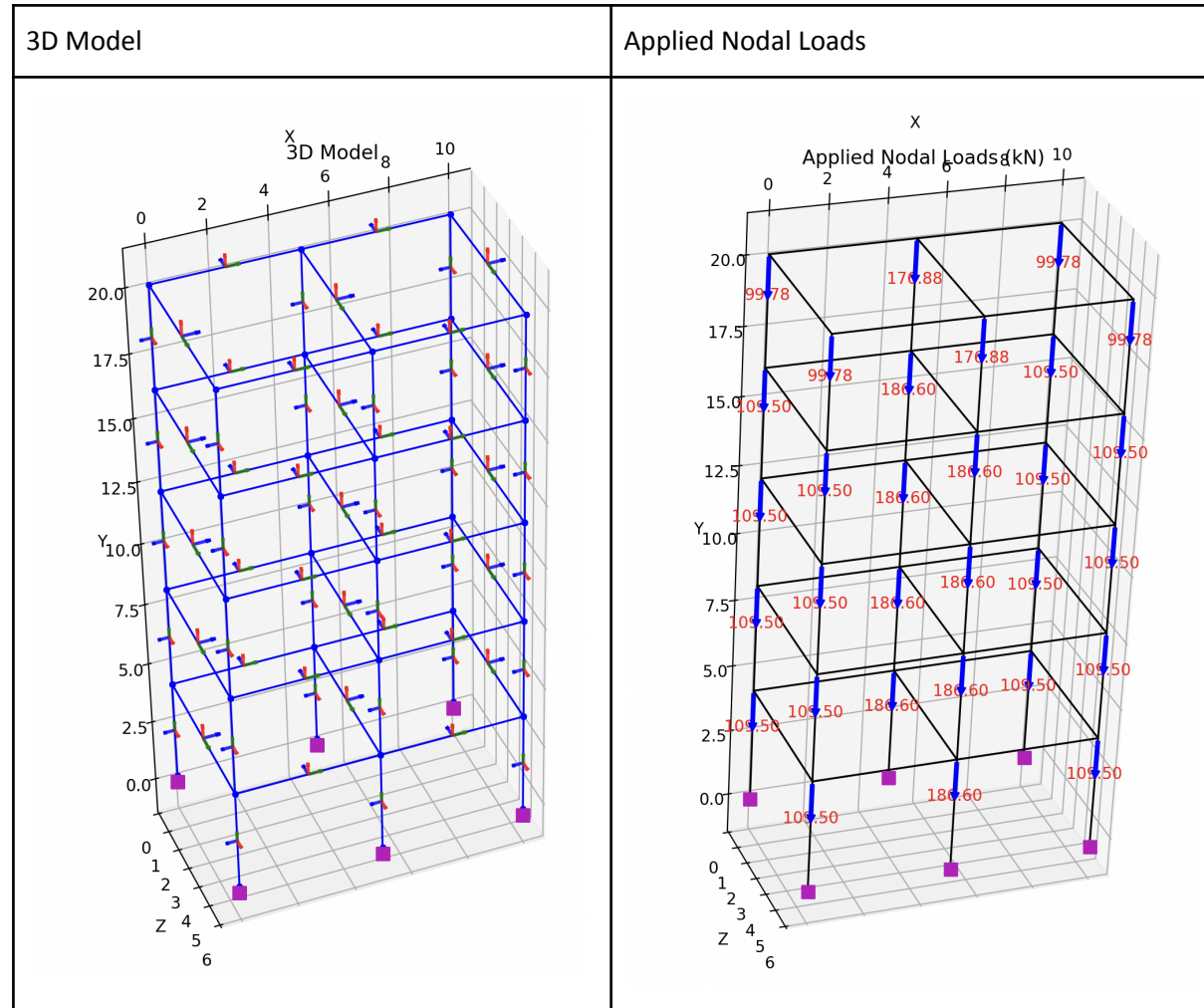
Code: https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_1_Assignment.py

Time Periods:

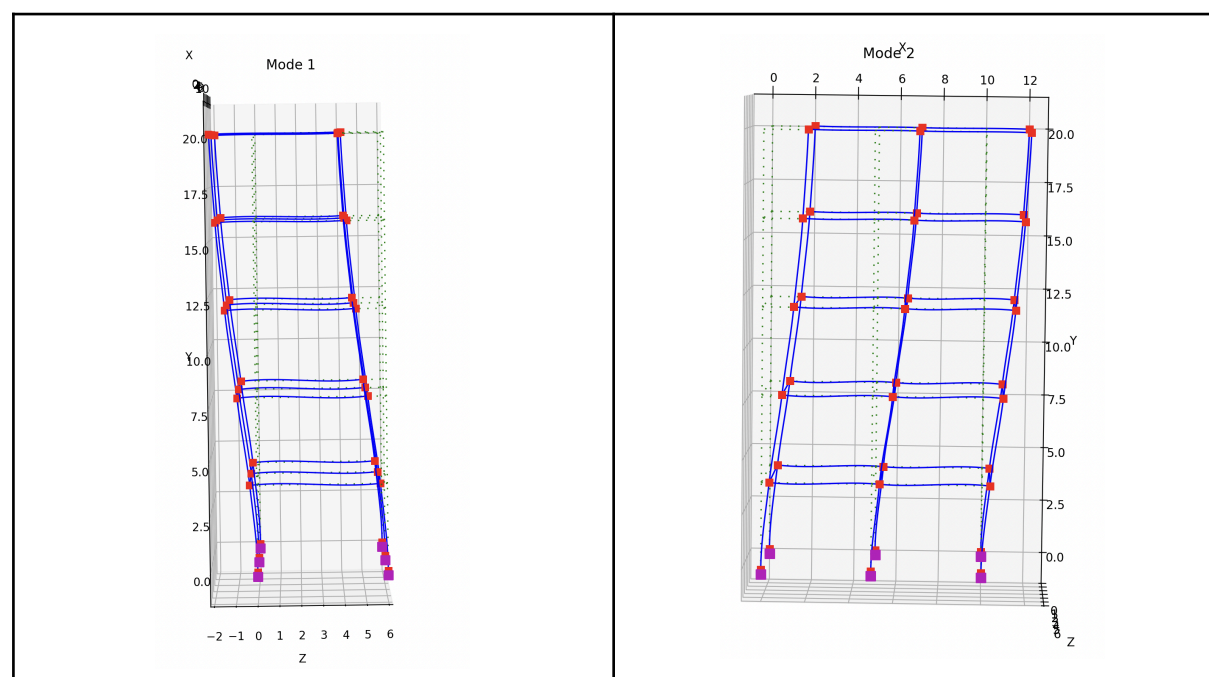
$T_1 = 1.09643\text{s}$
 $T_2 = 0.984784\text{s}$
 $T_3 = 0.863444\text{s}$
 $T_4 = 0.338194\text{s}$
 $T_5 = 0.307721\text{s}$
 $T_6 = 0.274183\text{s}$

Total Vertical Reaction = 3997.68 kN

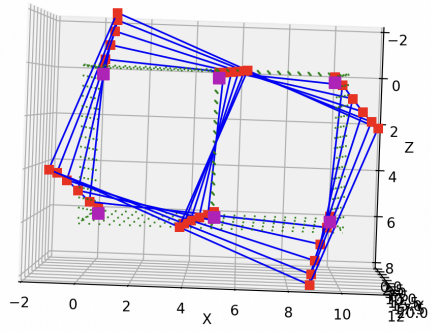
Diagrams:



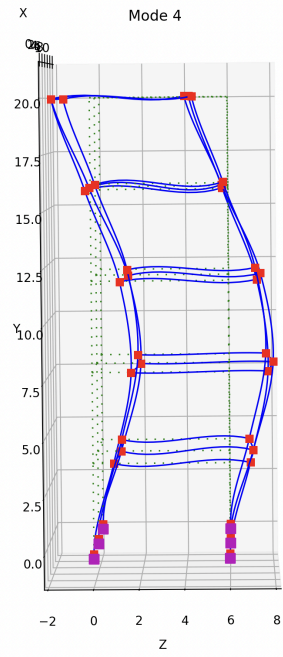
Mode Shapes:



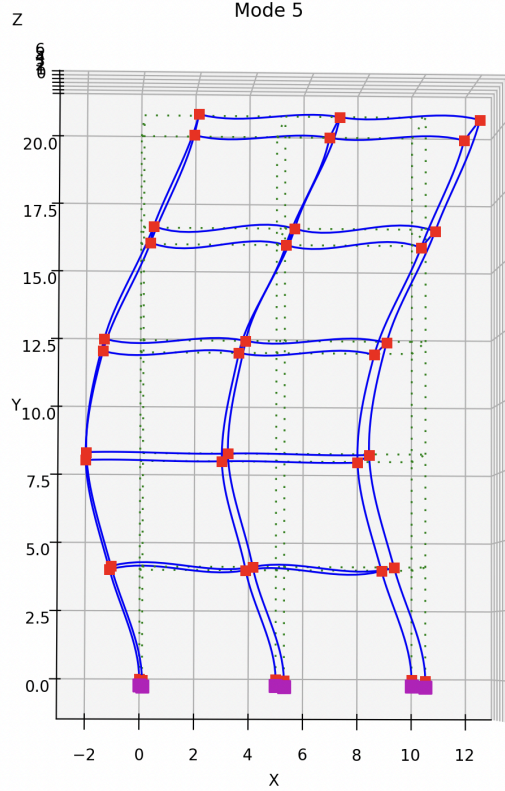
Mode 3



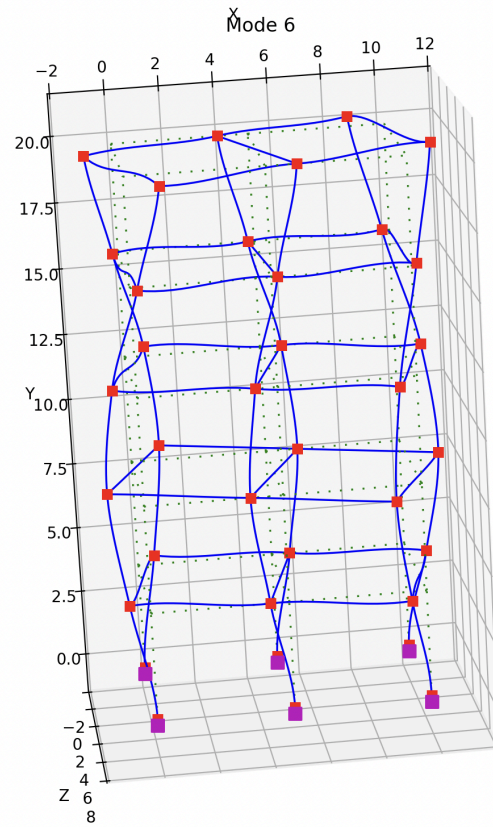
Mode 4



Mode 5



Mode 6



Session 2 Assignment (Moment Curvature Analysis)

Assignment

- Moment-curvature analysis
 - Monotonic loading
 - Develop moment-curvature diagram
 - Displacement control integrator
 - Load control integrator
 - Cyclic loading
 - Develop moment-curvature diagram
 - Try different steel and concrete materials
 - Try different nonlinear analysis algorithms
 - Try different convergence tests and tolerances

1) Monotonic Loading:

Displacement control integrator

Code: https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_2_Monotonic.py

Locking the axial DOF forces the axial strain at the centroid to be exactly zero. As a result, the cross-section is prevented from undergoing its natural axial expansion during bending. To satisfy this constraint, OpenSees develops a significant artificial compressive axial force within the section. This compression increases the load carrying capacity of the concrete fibers which can be seen in Figure 1,

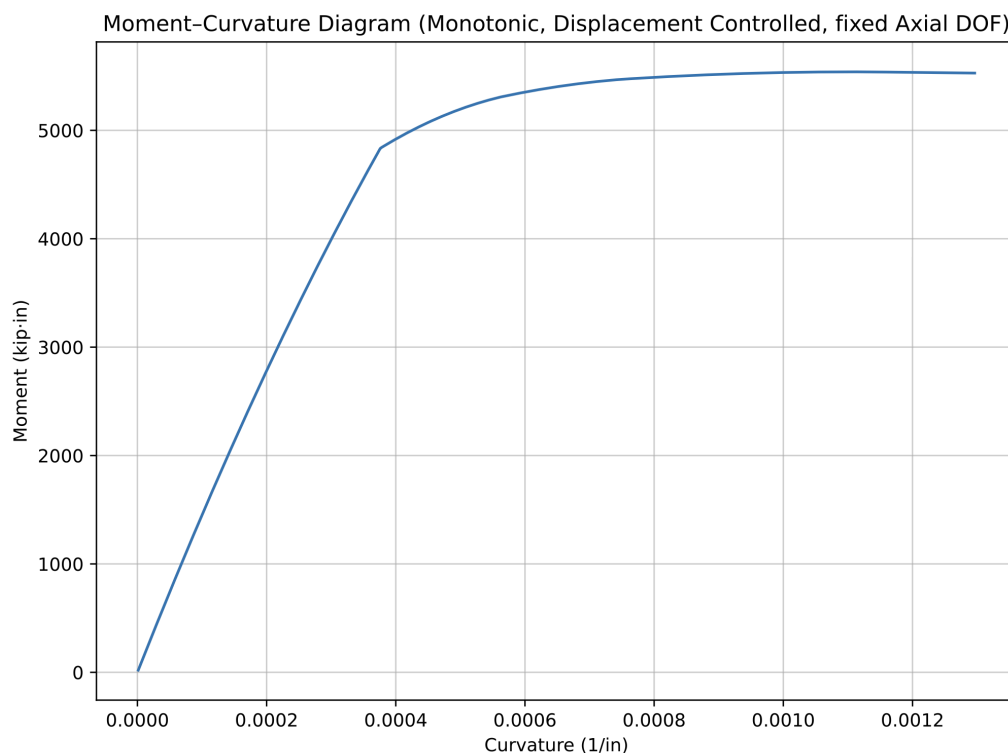


Figure 1. Moment Curvature Diagram (fixed Axial DOF), using Steel 01 & Concrete 01

This doesn't match the intended curve. Upon releasing the axial DOF allows the section to deform freely in the axial direction. This ensures that the resultant axial force across the fiber cross-section remains effectively zero. Under this condition, the computed response matches the accurate moment curvature analysis curve which can be seen in Figure 2,

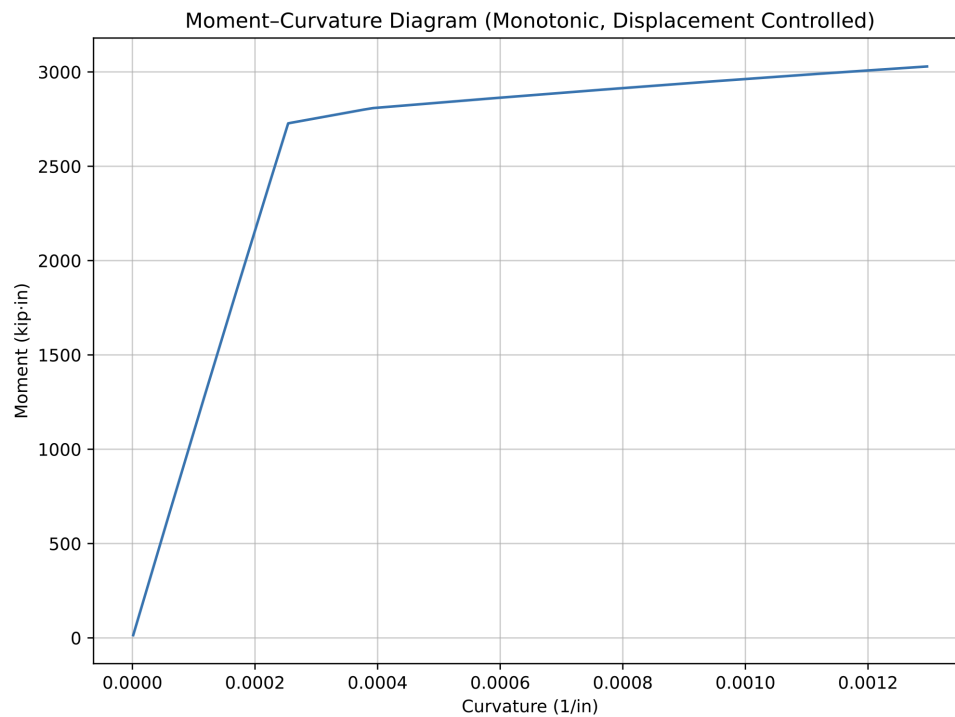


Figure 2. Moment Curvature Diagram (released Axial DOF), using Steel 01 & Concrete 01

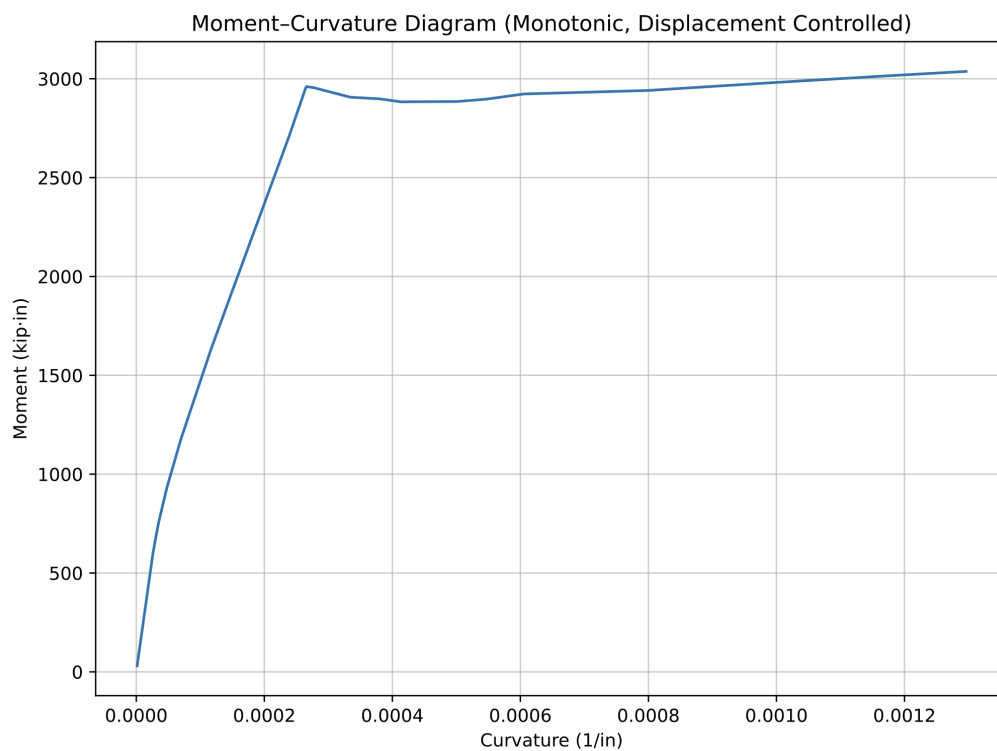


Figure 3. Moment Curvature Diagram (released Axial DOF), using Steel 01 & Concrete 02

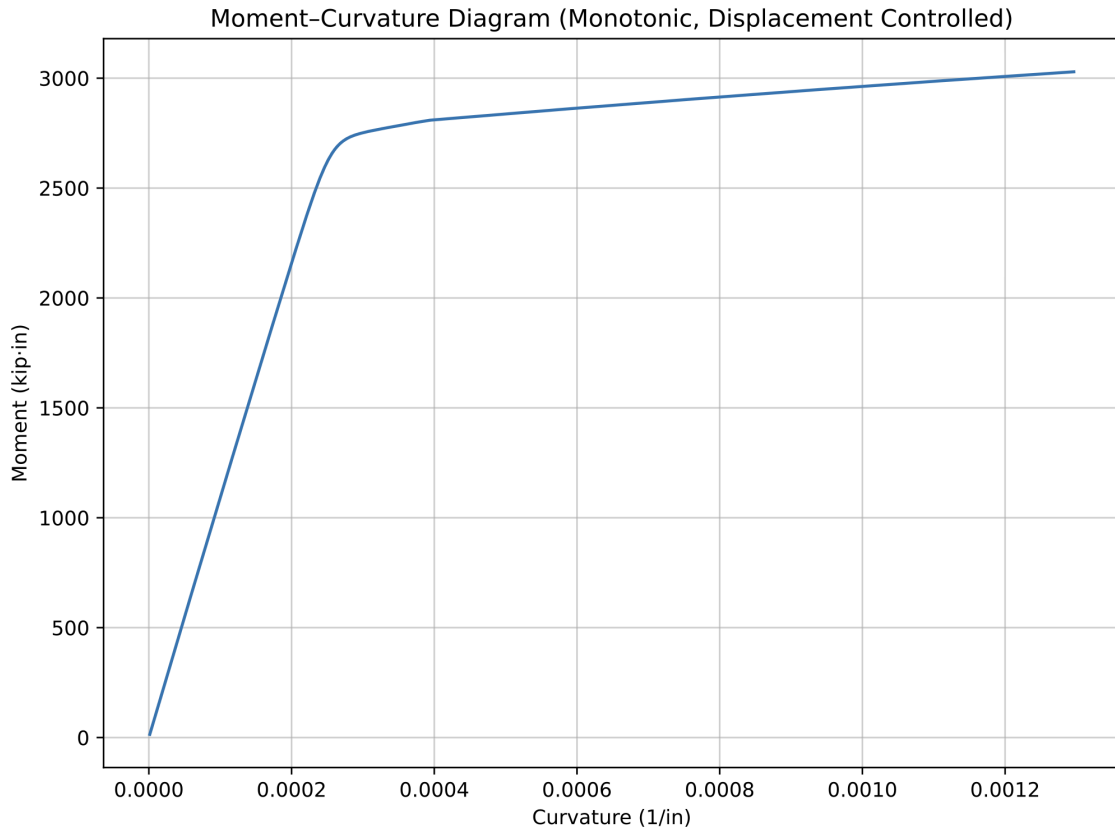


Figure 4. Moment Curvature Diagram (released Axial DOF), using Steel 02 & Concrete 01

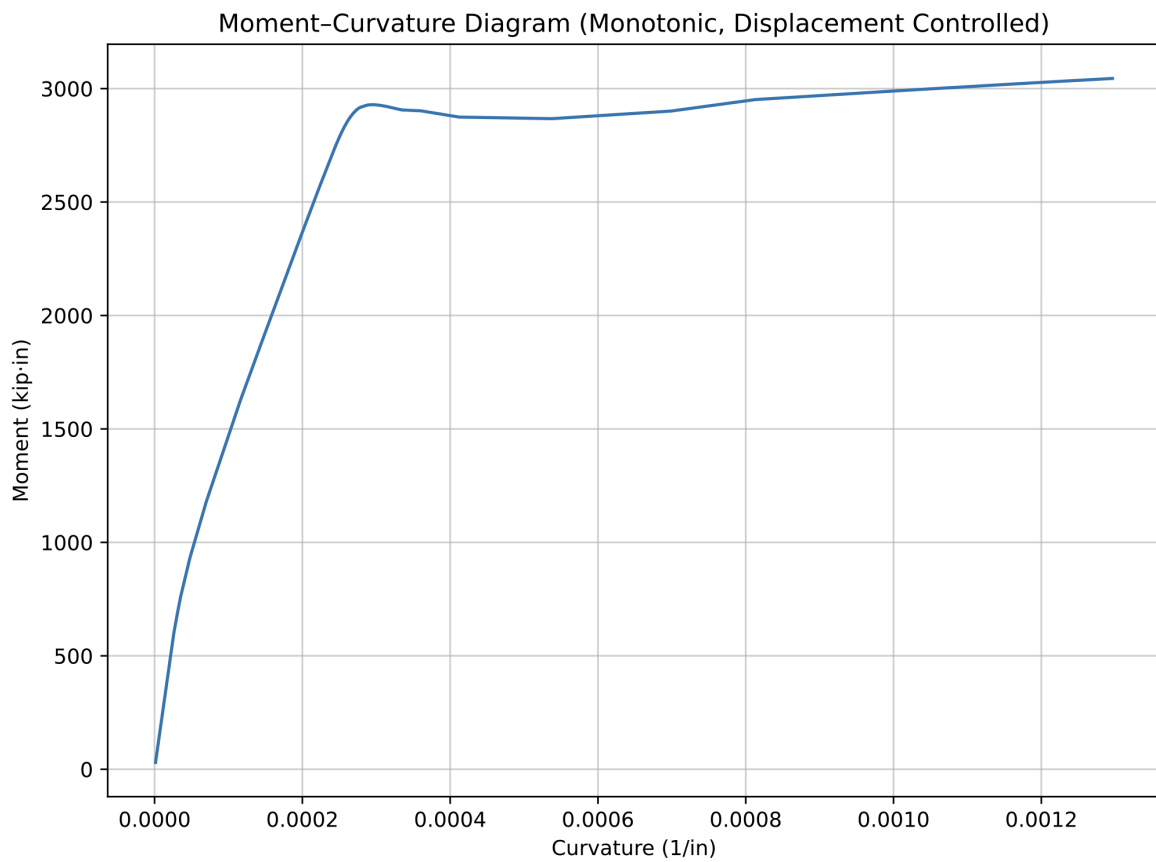


Figure 5. Moment Curvature Diagram (released Axial DOF), using Steel 02 & Concrete 02

Load control integrator

Code:

https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_2_Monotonic_Load_Controlled.py

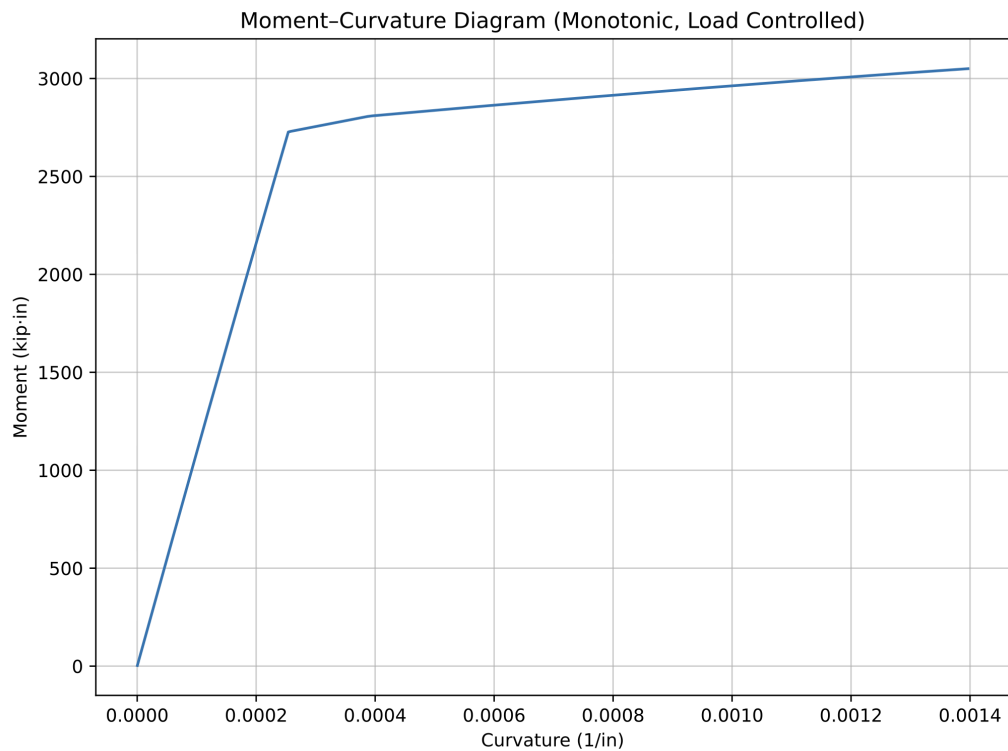


Figure 6. Moment Curvature Diagram (Load Controlled), using Steel 01 & Concrete 01

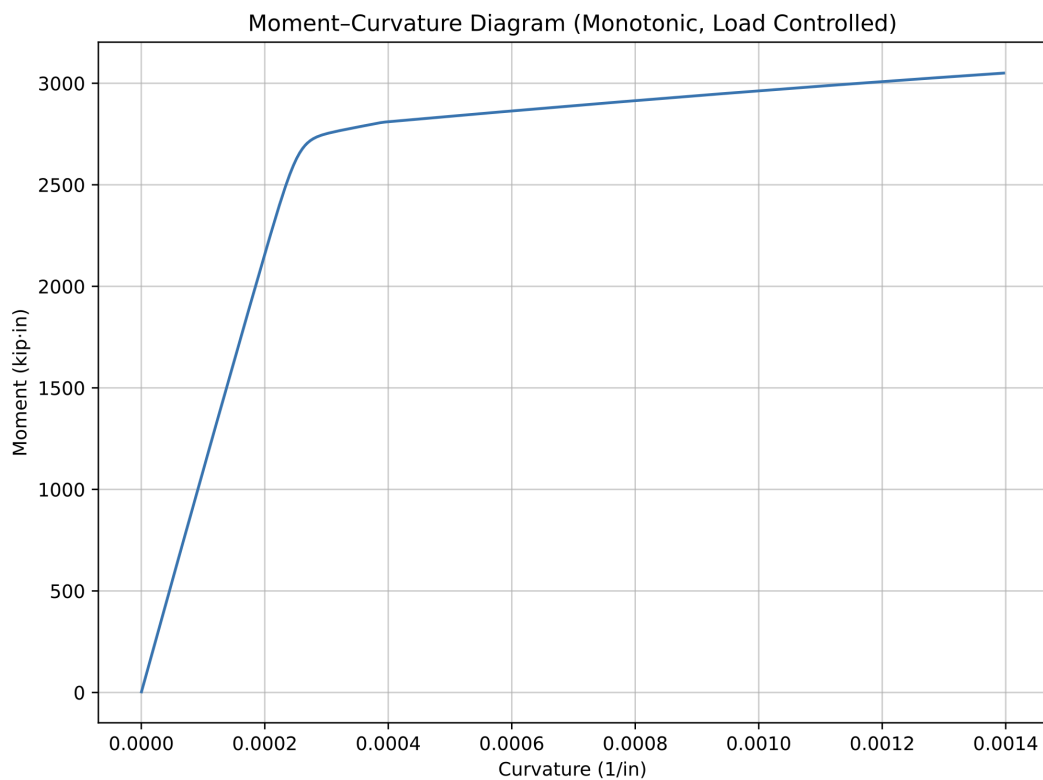


Figure 7. Moment Curvature Diagram (Load Controlled), using Steel 02 & Concrete 01

2) Cyclic Loading:

Code: https://github.com/nirajkyadav/Homeworks/blob/main/Tutorial/Session_2_Cyclic.py

Using Different Steel and Concrete Materials

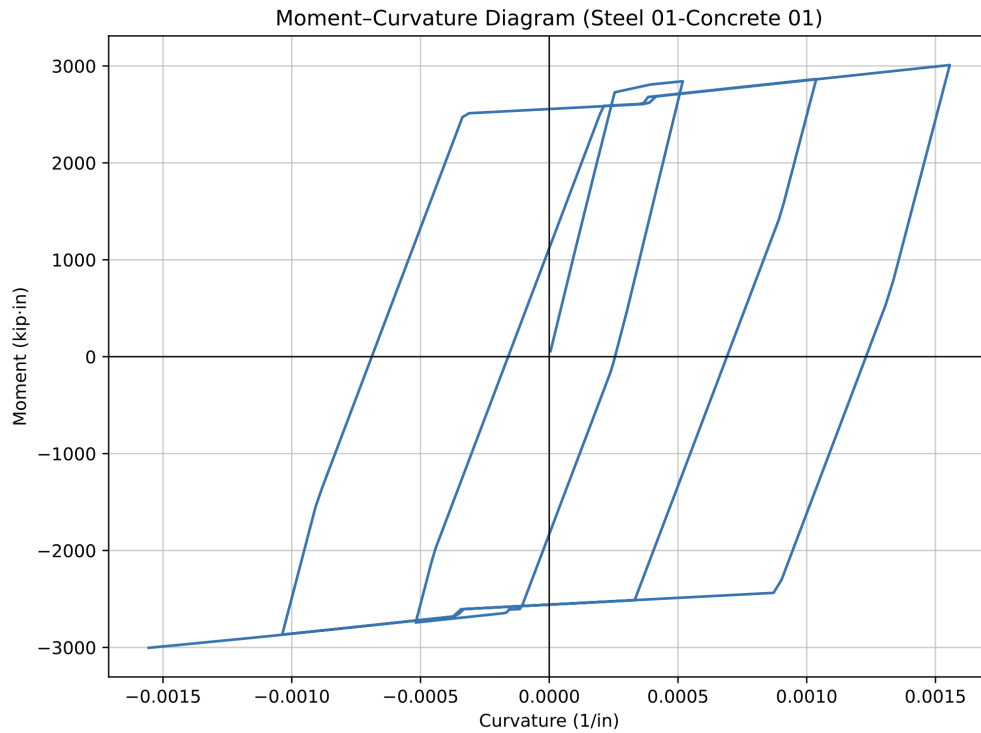


Figure 8. Moment Curvature Diagram, using Steel 01 & Concrete 01

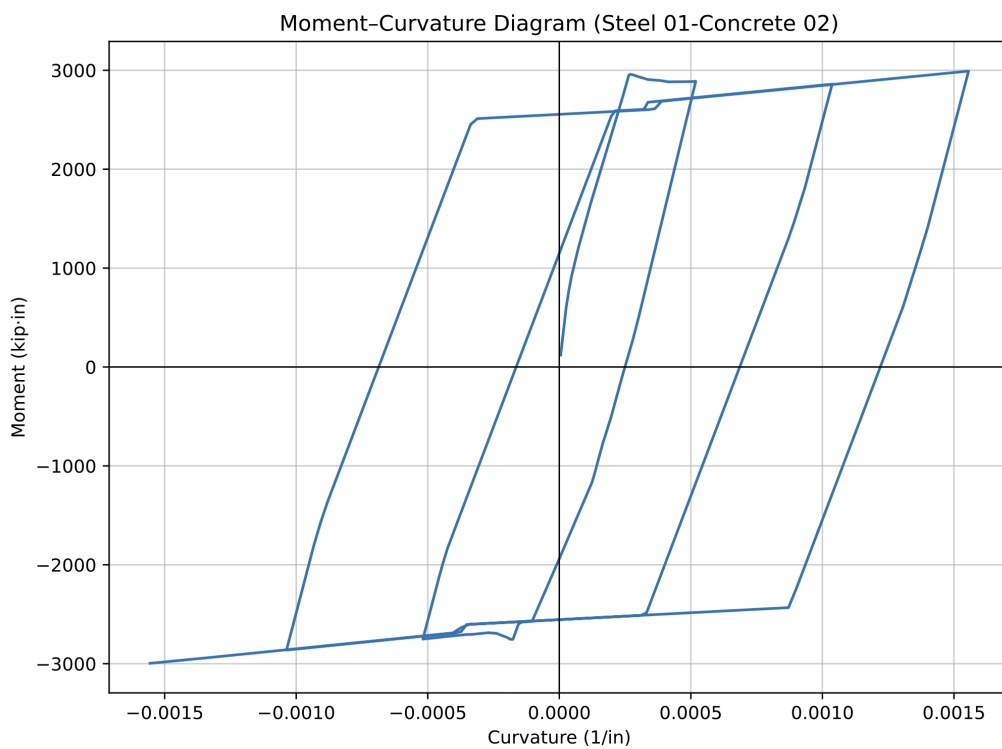


Figure 9. Moment Curvature Diagram, using Steel 01 & Concrete 02

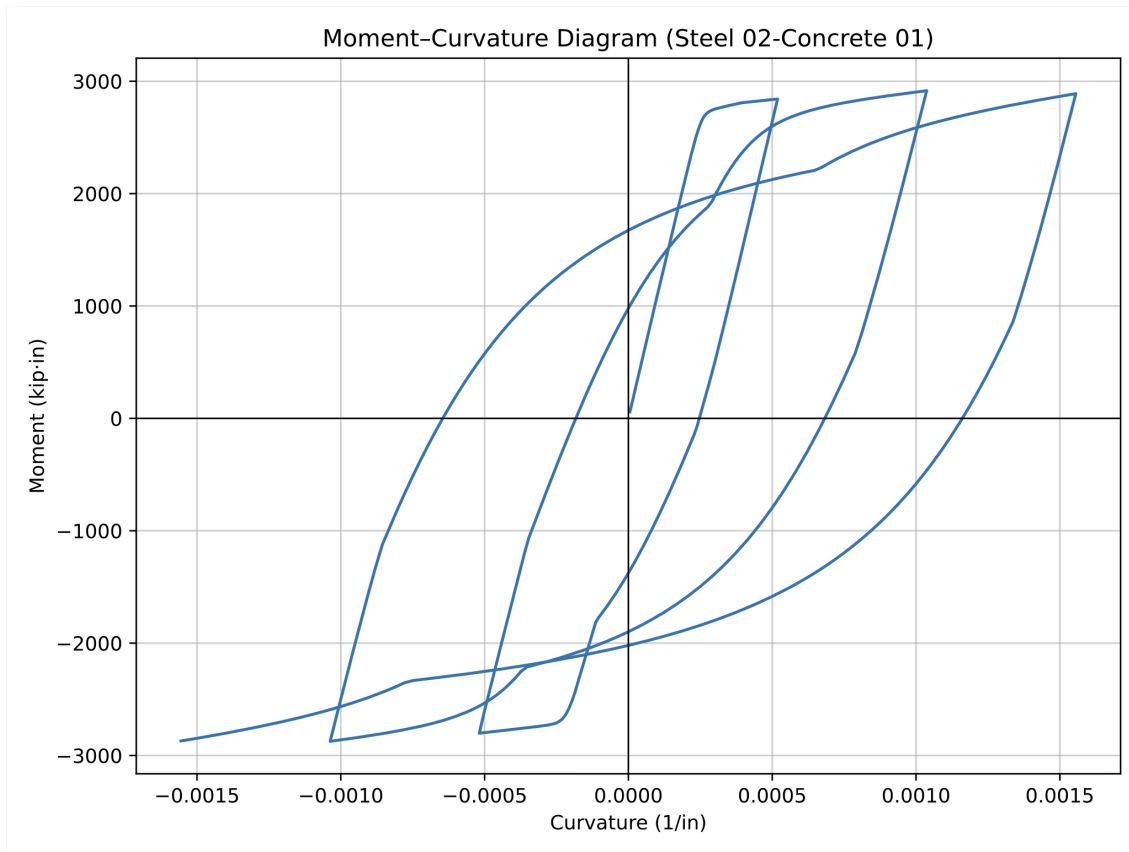


Figure 10. Moment Curvature Diagram, using Steel 02 & Concrete 01

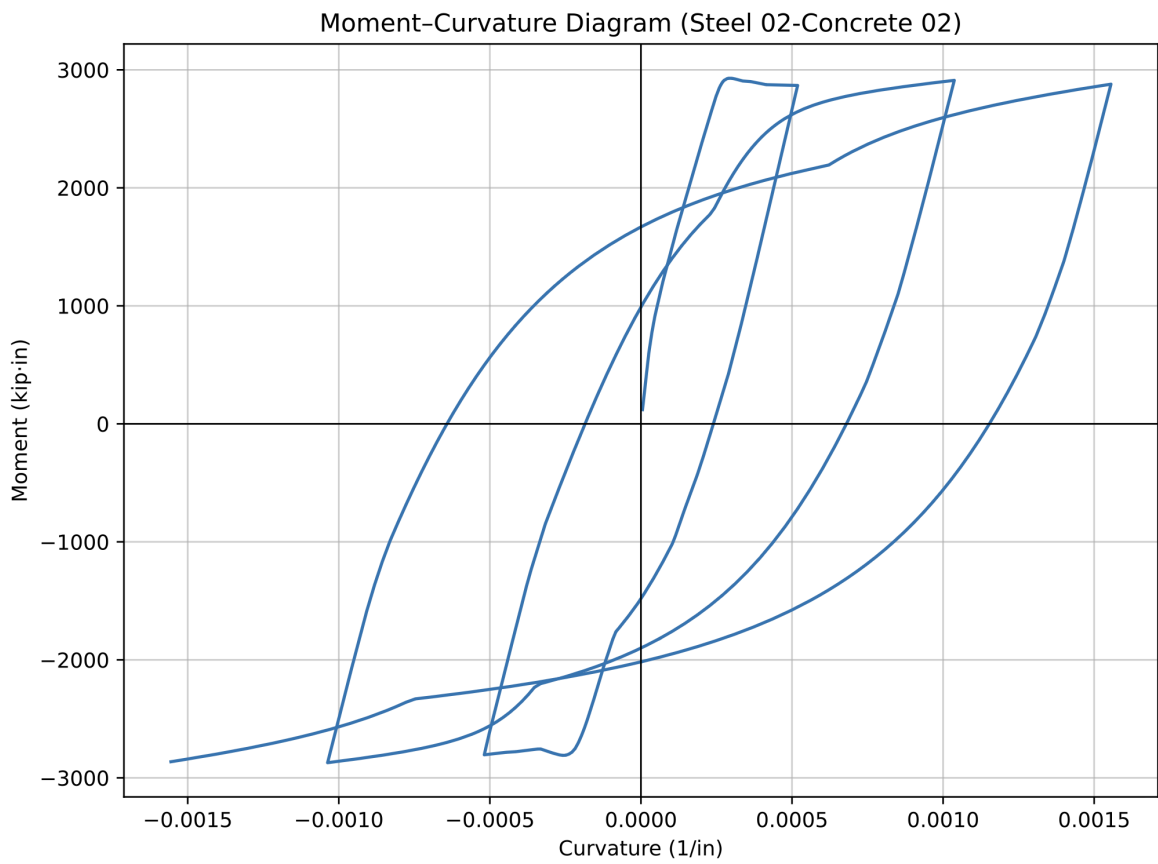


Figure 11. Moment Curvature Diagram, using Steel 02 & Concrete 02

Using Different Nonlinear Analysis Algorithm

Using Modified Newton the following error was encountered,

“

WARNING: CTestNormUnbalance::test() - failed to converge

after: 50 iterations current Norm: 0.990025 (max: 0.0001, Norm deltaX: 2.06957e-06)

ModifiedNewton::solveCurrentStep() -the ConvergenceTest object failed in test()

StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor -2583.67

OpenSees > analyze failed, returned: -3 error flag

Analysis failed at increment 75

”

This yield a wrong Curve (Figure 12),

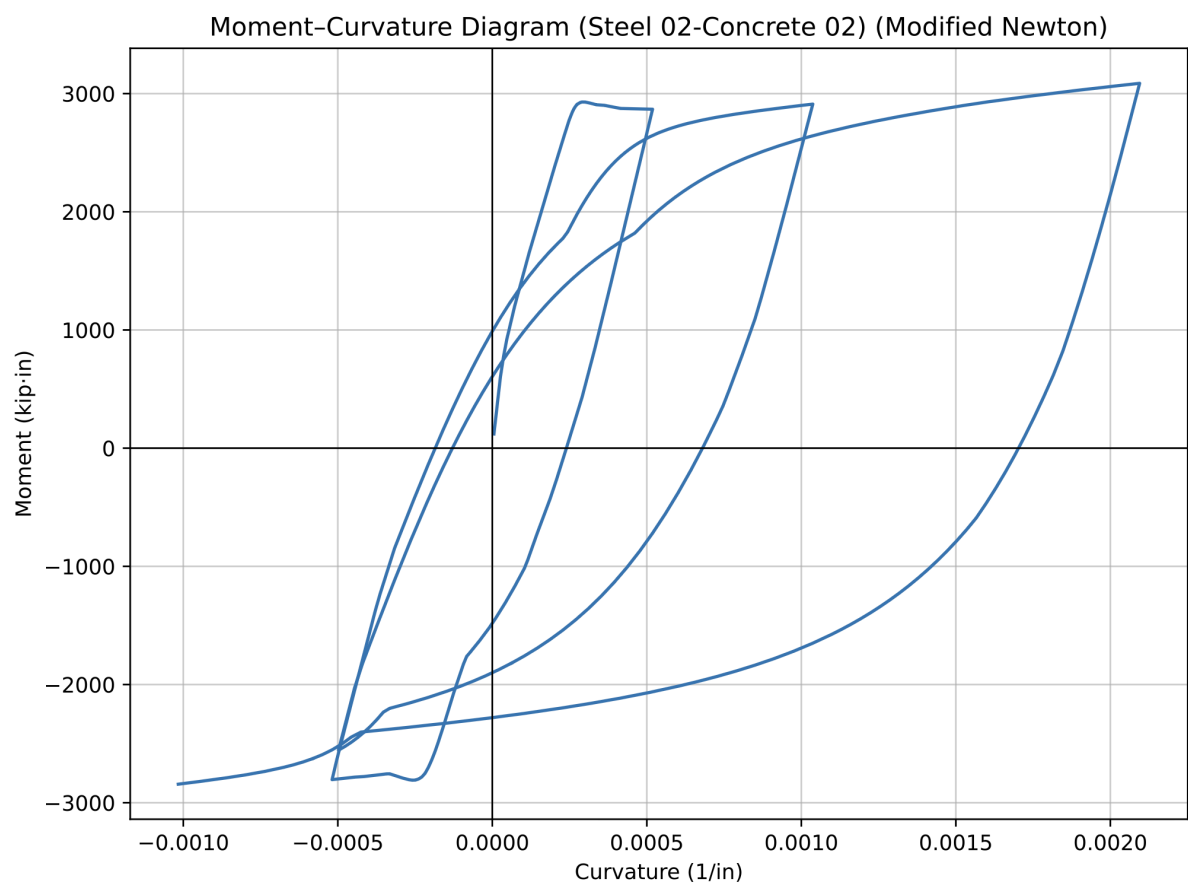


Figure 12. Moment Curvature Diagram (Modified Newton Algorithm), using Steel 02 & Concrete 02

Using Different Convergence Tests and Tolerances

Using RelativeNormDisplncr the following error was encountered,

“

WARNING: CTestRelativeNormDisplncr::test() - failed to converge
after: 50 iterations

NewtonRaphson::solveCurrentStep() -the ConvergenceTest object failed in test()

StaticAnalysis::analyze() - the Algorithm failed at step: 0 with domain at load factor 2615.33

OpenSees > analyze failed, returned: -3 error flag

Analysis failed at increment 2

.... load factor 2735.34

.... load factor 3174.61

.... load factor 2524.78

OpenSees > analyze failed, returned: -3 error flag

Analysis failed at increment 2

”

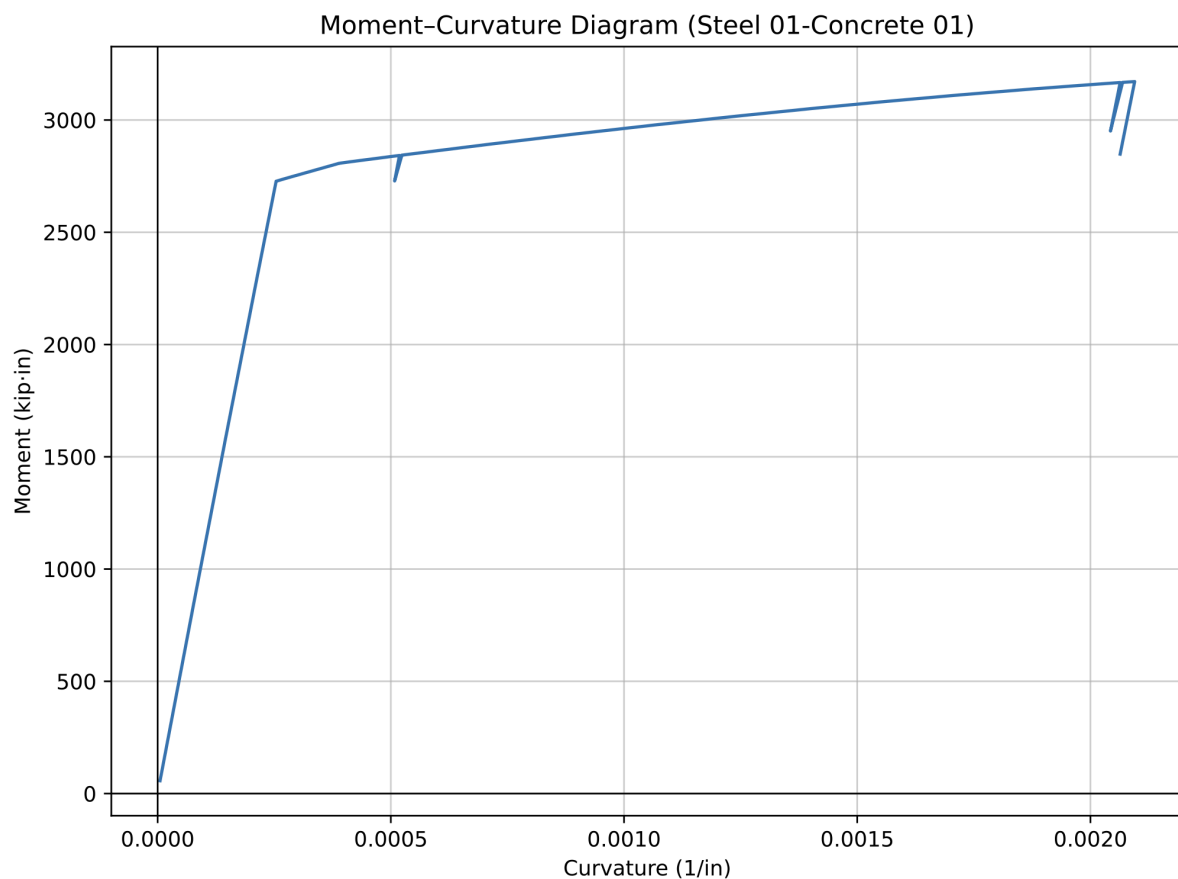


Figure 13. Moment Curvature Diagram (RelativeNormDisplncr), using Steel 01 & Concrete 01

Session 3, Hysteresis Curve Plot

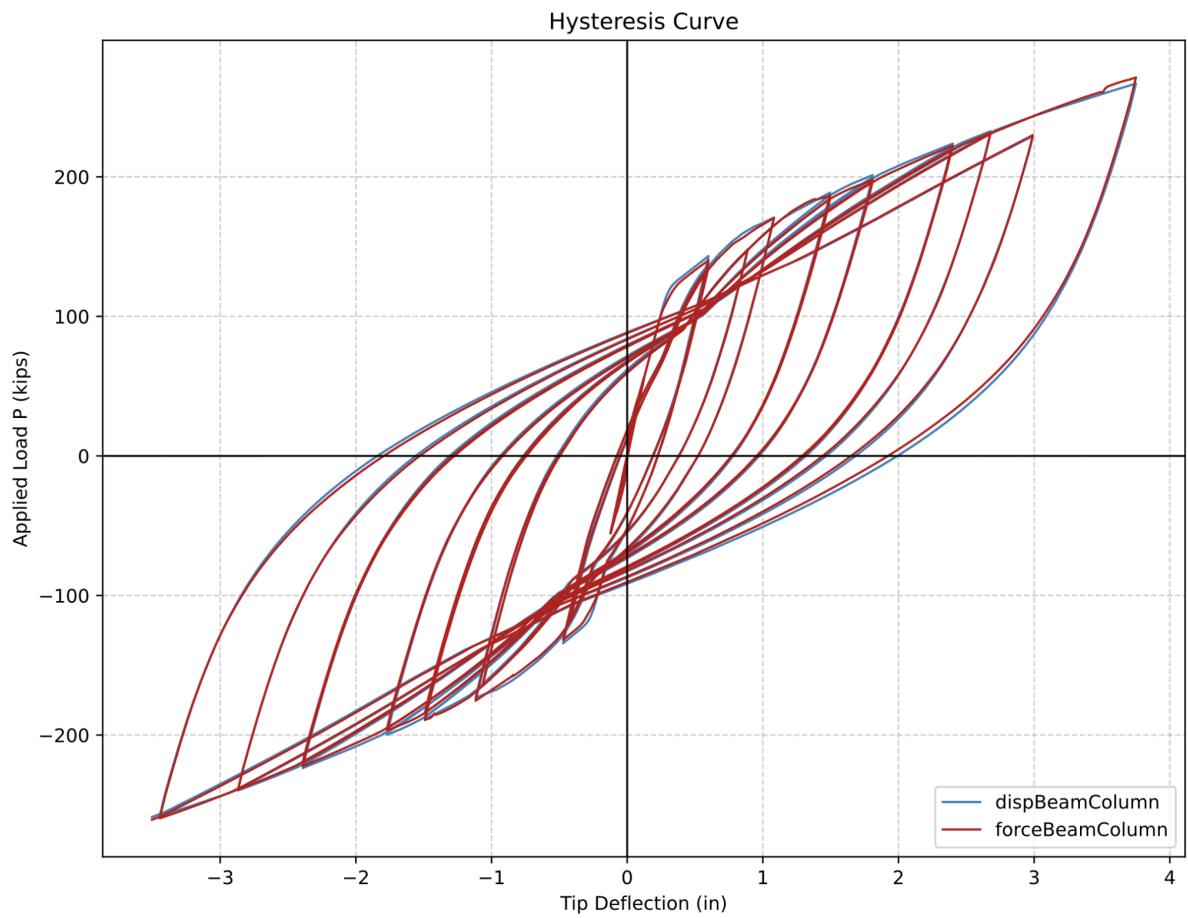


Figure 14. Hysteresis Curve Plot dispBeamColumn vs forceBeamColumn